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Practical evaluation of domestic airport Delay Tolerant Networking and its utilization

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Abstract

Recently, the Internet is an important infrastructure that supports the information and communication society. However, when a failure occurs due to a disaster, cables and wireless connections are cut off, resulting in a situation that people cannot communicate for a long period of time. Delay Tolerant Networking (DTN) is a method that allows information exchange even in situations that such networks are unavailable. Various implementations of DTN have been proposed so far. This paper first provides an overview of existing research on the airport DTN. One of the challenges for airport DTN so far is the evaluation of relatively short domestic flight routes. We evaluated the transmission efficiency when implementing DTN between Japan's domestic airports. The results show that it is effective not only in disasters but also in normal use.

Keywords: Type your keywords here, separated by semicolons ;

1. Introduction

Recently, the Internet is an important infrastructure that supports the information and communication society. However, when a failure occurs due to a disaster, cables and wireless connections are cut off, resulting in a situation that people cannot communicate for a long period of time. Delay Tolerant Networking (DTN) is a method that allows information exchange even in situations that such networks are unavailable [1]. DTN does not require a direct connection between the transmitting and the receiving point. It retains the data at the relay point, relays the data as

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soon as communication becomes possible, and finally transfers the data to the receiver. DTN routes data by stored and forward manner. Although this method takes time for communication, it has the potential to improve usage efficiency and reduce costs.

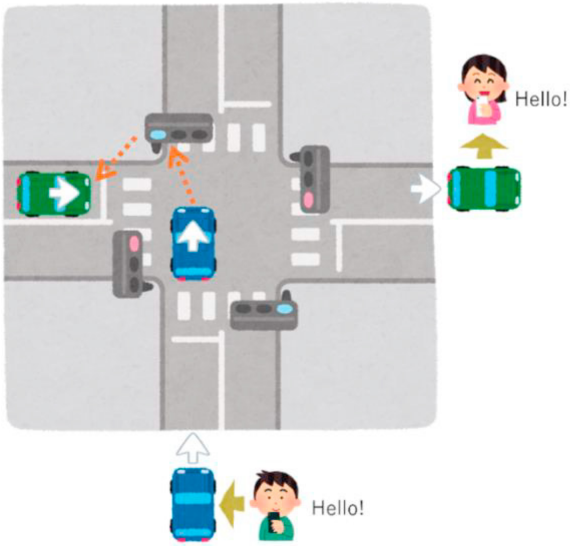

Fig. 1. Concept of DTN.

Various implementation forms of DTN have been proposed so far. DTN using cars is widely known [2], but recently DTN using drones has been actively proposed [3, 4]. On the other hand, one of the more practical methods is DTN, which uses airports as hubs and is operated by aircraft passengers. We call this the airport DTN. Since a relatively large number of passengers are on board an aircraft, each passenger retains communication data and the airport DTN enables large-capacity communications. Therefore, it may communicate more efficiently than a normal network by transmitting a certain amount of data or more.

This paper first provides an overview of existing research on the airport DTN. One of the challenges for airport DTN so far is the evaluation of relatively short-distance domestic flight routes. In this paper, we comprehensively evaluated the transmission efficiency when implementing DTN between Japan's Fukuoka Airport and other short-distance domestic airports. The results show that it is effective not only in disasters but also in normal use under certain conditions.

2. Existing research and issues on airport DTN

Martínez-Vidal et al. provided two different mobility models, a single day over the Atlantic Ocean and varying number of nodes over small oceans [5]. They assumed a direct contact between two airplanes flying in some directions. When contact duration of 390s and the airplanes spends a proportional amount of time, they estimated a average datalink of 14.72 Mbps. According to the results, they realized that it was possible to build simple airport DTN over Atlantic Ocean. Mallion et al. proposed a probabilistic routing algorithm called Contact Graph Routing for airport DTN [6]. Keraenen et al. showed through simulations that it is possible to build DTN communication on long-distance international flights using airports as relay nodes [7]. It is particularly targeted at connections between long-distance international airports. They use regular flights for three days and select flights from the middle of the week for calculations to avoid large differences in schedules. Specifically, they compared Delay, RTT, TCP throughput, DTN throughput, and Break Even using Helsinki Airport as the departure point and multiple European and US airports as destinations. A comparison table is shown in Table 1. Destination airports include Stockholm, Frankfurt, Madrid, New York and Los Angeles.

In Table 1, Destination is the arrival airport, which is about 1,540 km from Helsinki to Frankfurt. Other airports are Madrid, approximately 2,940 km away, New York, approximately 6,600 km, and Los Angeles, approximately 9,000 km. Flight Time is flight time between airports. Delay is delaying time. RTT is Round Trip Time. TCP Throughput is communication speed when using TCP communication on the Internet among these airports. DTN Throughput is communication speed when using airport DTN. Break Even is DTN communication speed that refers to the amount of data faster than TCP communication.

Table 1. Evaluation of long-distance airport DTN.

Destination	Flight Time (Min.)	Delay (Sec.)	RTT (ms)	TCP throughput (Mbps)	DTN throughput (Mbps)	Break Even (GB)
Stockholm	60	10800	8	40.0	333	54
Frankfurt	180	14400	40	8.0	250	14
Madrid	240	21600	60	5.3	167	14
New York	480	36000	120	2.7	100	12
Los Angels	720	50400	190	1.7	71	11

This study mainly examines airports in Europe and United States. The target airports are long-haul international airports, and verification has not been conducted at short-haul domestic airports that are frequently used daily. Therefore, we evaluate the practicality of short-haul domestic airport DTN in Japan and consider how to utilize them effectively.

3. Practically evaluation of the domestic airport DTN

3.1. Airport DTN model

Airports function as communication start and end nodes. Transfer airports function as transit nodes. Data is transferred between airports by keeping the data on aircraft passengers. Airports have the function of hubs. Passengers have a data relay function when moving from one aircraft to another, and a local storage function for passengers waiting for a connection. Passengers aboard aircrafts at departure airports and leave their mobile devices powered on. Nowadays, passengers often store electronic tickets on their mobile devices. They contain data on the passenger's route and final destination [8]. While passengers wait to board, mobile devices can exchange route information and data. Then all passengers only keep messages to their final destination. If there are messages for a destination with no passengers, a passenger on another aircraft will carry these data and eventually the data will be passed to the passenger heading to the destination. Fig.2 shows the concept of airport DTN.

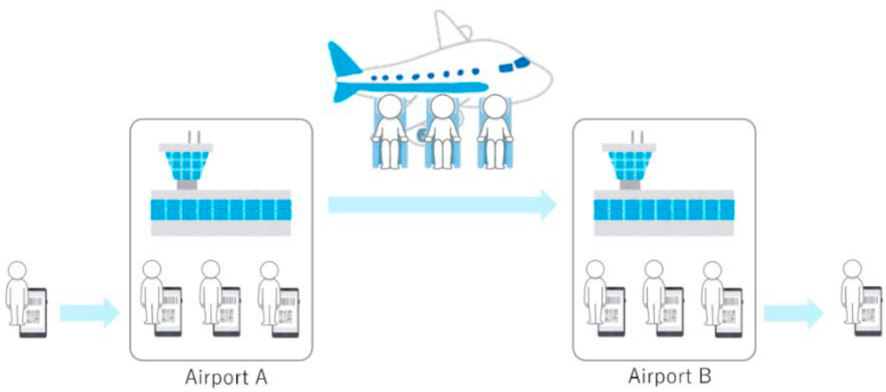

Fig. 2. Concept of airport DTN.

Passengers' ability to transfer data is determined by their mobile device local storage capacity. The simulation assumes that passengers stay at the airport for one hour before the aircraft departs and one hour after the aircraft lands. Each passenger has 1GB of data storage capacity. Boeing 747-400 and Airbus 340-600 have approximately 400 seats, so each aircraft can transfer 400GB.

In the evaluation, we first redefine each item in Table 1. Then we make and compare the domestic airport DTN evaluation table.

3.2. Evaluation indexes

Firstly, Destination is the distance between airports. Since this research is conducted within Japan, we will use Fukuoka Airport as the starting point, 211 km to Miyazaki, 449 km to Kansai Airport, 476 km to Itami Airport, and 599 km to Chubu Airport. Secondly, Flight Time is calculated from the distance between airports. Starting from Fukuoka Airport, it takes 40 minutes to Miyazaki Airport, 70 minutes to Kansai Airport, 65 minutes to Itami Airport, and 75 minutes to Chubu Airport. Delay is the time from when data download starts at the departure airport until it completes uploading at the arrival airport. Passengers will spend one hour transmitting at the departure airport and one hour receiving at the arrival airport. Total time of transmitting and receiving is two hours. Delay is the total time plus the flight time. Round Trip Time (RTT) is the round-trip time of communication from sending data to the communication partner until receiving a response. We investigate the servers of airport management companies near each airport to calculate RTT, and measured transmission time of ICMP packets from Fukuoka to servers in various locations. Table 2 shows information about each server. We were unable to find any servers near airports other than these three servers. Therefore, we predict RTT of other five airports by a regression analysis using those three RTT values and the distance between airports. The analysis is calculated by using Equation (1), where x is the location of the airport and y is RTT value.

Table 2. RTT to servers near each airport.

Airport	Server Name	Location	RTT average (ms)
Shin-Chitose	S.T.E.P Sapporo Data Center	Ebetsu, Hokkaido	33
Itami	Narita International Airport Corp.	Ibaraki, Osaka	13
Haneda	Airport Facilities Co. Ltd.	Shinjuku, Tokyo	19

$$y = ax + b, \qquad a = \frac{\sum\{(x_n - \bar{x})(y_n - \bar{y})\}}{\sum(x_n - \bar{x})^2}, \qquad b = \bar{y} - a\bar{x} \tag{1}$$

Next, TCP Throughput is the communication speed when using TCP on the Internet. As shown in Equation (2), it is determined by the window size, which is the buffer capacity when transmitting and receiving data, and RTT for the round trip of packets between communication hosts. The window size is commonly assumed to 64KB.

$$TCP\ Throughpuut = \frac{Window\ size}{RTT} \tag{2}$$

DTN Throughput is the communication speed when using DTN and is calculated using Equation (3).

$$DTN\ Throughpuut = \frac{Total\ data}{Delay} \tag{3}$$

Break Even is the break-even point for DTN and TCP. This refers to the amount of data at the boundary that transfer using DTN is faster than transfer using TCP. This is determined by the performance of TCP communication as shown in Equation (4).

$$Break\ Even = TCP\ Throughput \times Delay \tag{4}$$

3.3. Evaluation results

Table 3 shows DTN evaluation results calculated according to the evaluation indexes. Compared to Table 1 of existing research, the overall flight time is shorter due to the short distance among airports, and Delay and RTT are also shorter than previous one accordingly. On the other hand, TCP Throughput and DTN Throughput have increased significantly. Regarding TCP Throughput, one of the reasons is the shortening of RTT. The improvement in DTN Throughput is due to an increase in the amount of transmission per unit time with a shorter flight time. Break Even is also increased as TCP Throughput improves. In other words, the break-even point at which DTN becomes effective is higher than for long-haul flights. This means that the effectiveness of DTN cannot be demonstrated unless a larger amount of data is sent.

Table 3. DTN comparison in case of Japan domestic airports.

Destination	Flight Time (Min.)	Delay (Sec.)	RTT (ms)	TCP Throughput (Mbps)	DTN Throughput (Mbps)	Break Even (GB)
Miyazaki	40	9600	5.26	97.34	375.0	116.81
Itami	65	11100	13	39.38	324.32	54.64
Kansai	70	11400	14.44	35.46	315.79	50.53
Chubu	75	11700	15.97	32.06	307.69	46.89
Haneda	85	12300	19	26.95	292.68	41.44
Naha	90	12600	20.56	24.90	285.71	39.22
Narita	105	13500	25.15	20.36	266.67	34.36
Shin-Chitose	130	15000	33	15.52	240.0	29.09

4. Airport DTN practicality evaluation

We investigated the daily cumulative traffic volume and time-series changes for each airport in order to evaluate the practicality of airport DTN. Firstly, we compared TCP and DTN in terms of cumulative daily traffic for each airport. The cumulative traffic volume per day at each airport was calculated from the flight schedule for each airport [9]. An example flight schedule is shown in Table 4, and the comparison results are shown in Fig. 3. The cumulative traffic of TCP is inversely proportional to the distance according to Fig. 3. This shows that DTN is superior to all airports except Miyazaki Airport and indicates that DTN has a high value of use. DTN is particularly effective for the Fukuoka Haneda flight, since it has many passengers.

Table 4. Example of daily flight schedule for Fukuoka Haneda flight.

Airline Company*	Departure Time	Arrival Time	Arrival Airport**	Airplane	Seats Num.	DTN Throughput (GB)
SFJ	7:00	8:25	HND	A320	162	105
JAL	7:30	8:55	HND	A350-900	391	254
APJ	13:55	15:55	KIX	A320	180	126
ANA	17:50	18:55	ITM	B767-300	270	194
SFJ	20:25	21:45	NGO	A320	180	123
SKY	10:30	12:40	CTS	B737-800	177	94

*SFJ: Starflayer, JAL: Japan Airlines, APJ: Peach Aviation, ANA: All Nippon Airways, SKY: Skymark Airlines
**HND: Haneda Airport, KIX: Kansai Airport, ITM: Itami Airport, NGO: Chubu Airport, CTS: Shin-Chitose Airport

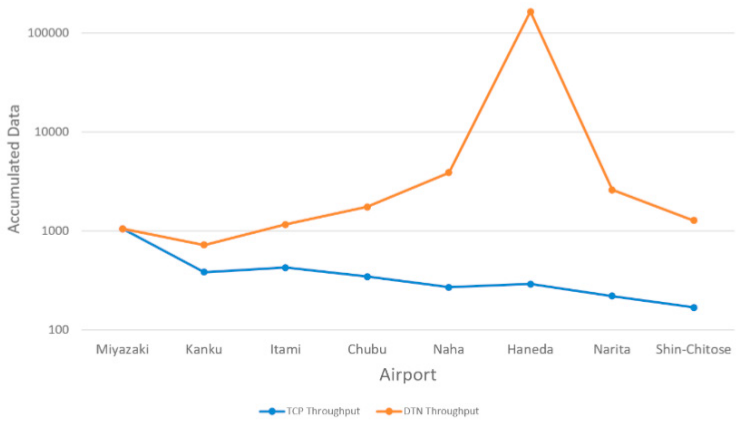

Fig. 3. Cumulative daily traffic for TCP and DTN.

Next, we compared the daily time-series cumulative traffic volume. Table 5 shows the evaluation results, and Fig.4 shows an example of the chronological changes in cumulative traffic. The DTN effectiveness rate indicates the percentage of time during the day when DTN has higher communication capacity. Since DTN communication is not available until the first flight, the overall value of using TCP is high. At Miyazaki Airport, which is a short distance away, TCP has higher transfer capacity at any time of day. DTN is often more effective by using TCP and DTN differently depending on the time of day at other airports.

Table 5. Evaluation of cumulative traffic over time.

Arrival Airport	Evaluations	DTN effectiveness rate (%)
Miyazaki	Since TCP always exceeds DTN, the value of using TCP is always high.	4
Chubu	DTN exceeds TCP at 12 o'clock. TCP is more useful until 12:00, and DTN is more valuable after 12:00.	54
Shin-Chitose	DTN exceeds TCP at 13:00. TCP is more useful until 13:00, and DTN is more valuable after 13:00.	46
Kansai	DTN exceeds TCP at 17:00. TCP is more useful until 17:00, and DTN is more valuable after 17:00.	38
Itami	DTN exceeds TCP when the first flight arrives. The value of using TCP is high until the arrival of the first flight, and DTN is high after the arrival of the first flight.	63
Haneda		63
Naha		63
Narita		63

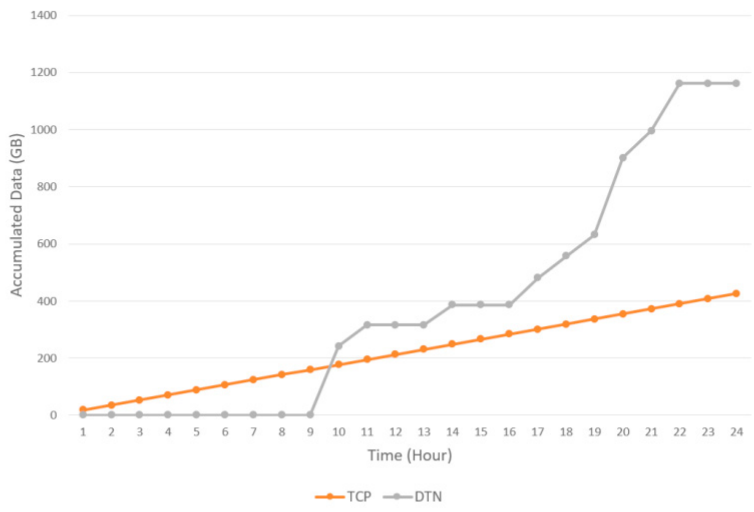

Fig. 4. Cumulative communication volume for Fukuoka Itami flights

5. Conclusion

This paper presented existing research on DTN using aircraft, called airport DTN. Since existing research only targets long-haul flights, we conducted verification on short-haul flights. The evaluation results showed that DTN needs to send a larger amount of data to be effective on short-haul flights compared to long-haul flights. Furthermore, when we calculated the daily cumulative communication volume, we realized that DTN was effective at all airports except for the one with the shortest distance. Looking at changes in daily data traffic, we found that the effective optimal time for DTN differs at each airport.

For further study, actual communication does not include DTN, but also Cellular and Wi-Fi networks. Such heterogeneous network environment can provide more stable communication infrastructure. We will investigate issues to provide such heterogeneous network and propose the solutions. The solutions may include Appropriate network selection according to application demands, link status and contact timing.

Generally, DTN is often considered to be used for special purposes such as during disasters, but if used properly, it can be used as a daily communication service. In the future, it is expected to be used as a means of communication in outer space, where it is difficult to install base stations [10].

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